

Proposed Solution

Date	15 July 2026
Team / Submission ID	
Project Name	Emotion Detection & Learning Support Engine
Submitted By	Venu Gopal (Sole Team Member / Team Lead)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This proposal aims to make online self-study emotionally aware by combining two complementary emotion-classification models with a generative AI layer, so that struggling or disengaged learners receive an empathetic, actionable response instead of silence.

Project Overview

Field	Details
Objective	Detect a learner's emotional state from free-text problem descriptions and respond with personalised, field-specific learning support, reducing silent disengagement.
Scope	A Streamlit web application covering text preprocessing, dual-model emotion classification (BiLSTM + BERT), mixed-emotion and keyword-based refinement, Gemini-generated responses with a rule-based fallback, interaction logging, and an analytics dashboard.

Problem Statement

Field	Details
Description	Learners struggling with a topic have no channel to signal how they feel, so platforms cannot intervene with the right kind of support at the right moment.
Impact	Left unaddressed, this contributes to disengagement, lower topic mastery, and higher drop-off from self-paced courses.

Proposed Solution

Field	Details
Approach	Preprocess the learner's text (NLTK cleaning/tokenising), run it through a Keras BiLSTM model and a fine-tuned BERT model in parallel, blend/boost the predictions with a keyword-based mixed-emotion detector, then hand the result to Gemini 2.5 Flash to draft a supportive, field-specific response — falling back to a curated rule-based response if Gemini is unavailable.
Key Features	- Side-by-side BiLSTM vs BERT emotion comparison with confidence scores - Mixed / secondary emotion detection - Gemini-generated personalised learning guidance with a safe fallback - CSV interaction logging and a Plotly analytics dashboard (emotion distribution, emotional journey, field-wise breakdown)

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing resources for model inference	Standard CPU (GPU optional for faster BERT inference)
Hardware	Memory	8 GB RAM minimum
Hardware	Storage	Local disk for model weights (~30 MB BiLSTM + BERT model files), datasets and CSV logs

Resource Type	Description	Specification / Allocation
Software	Frameworks	Streamlit (UI), TensorFlow/Keras (BiLSTM), PyTorch + Transformers (BERT)
Software	Additional libraries	NLTK, pandas, Plotly, scikit-learn, google-generativeai, python-dotenv
Software	Development environment	VS Code / Jupyter Notebook, Python 3.10+
Data	Training / reference data	eng_dataset.csv, emotion-emotion_69k.csv (archive), emotion_response_examples.csv (logged interactions)